

Black Hole Migration-Merger Dynamics in AGN Discs

Valentín Peña-Donaire¹ (vipena@uc.cl), Cristobal Petrovich², Julio Chanamé¹

¹Institute of Astrophysics, Faculty of Physics, UC Chile. Av. Vicuña Mackenna 4860, 7820436 Macul, Chile.

²Department of Astronomy, Indiana University Bloomington. IN 47405, USA.



1. Abstract

To this day, there is still no strong consensus on the theory behind neither the formation channel nor the mass distribution of the local population of intermediate-mass black holes (IMBHs). We evaluate the possibility of stellar-mass black hole (sBH) hierarchical mergers in the innermost parsec of active galactic nucleus (AGN) accretion discs, where gravitational interactions with it drive embedded black holes (BHs) to migrate. The migration dynamics of point-like bodies embedded in a gas disc has been extensively studied for protoplanetary discs, where annuli of convergent migration trap protoplanets. The structure of these migration traps depends on the disc radial profiles of scale height, temperature, surface density, and their gradients. The central body mass and accretion rate, and the Shakura-Sunyaev parameter α that characterises the magnetorotational instability viscosity of the disc, also play a role. Lastly, whether a trap halts the migration of a body or not depends on the mass ratio between the small body and the central object. Here, we numerically integrate the differential equations governing the migration of embedded bodies, adapted to an AGN accretion disc and the BH population embedded in it. The initial population synthesis follows from a Bahcall-Wolf cusp, modified so that BHs in the spherical nuclear star cluster fall onto the disc before migration. We derive the mass distribution of BHs that fall into the super-massive BH (SMBH), with implications for the gravitational wave (GW) signature of this process. In the future, these results can be used to estimate the mass distribution of BHs and their progenitors in the local Universe, testing against LISA observations.

2. BHs in AGN accretion discs

Accretion discs (ADs) in AGNi contain stars and stellar remnants [1]. These migrate radially, as do protoplanets in protoplanetary discs (PPDs) [2]. This process stagnates at annuli of convergent migration, dubbed “migration traps”, where sBHs undergo hierarchical merging until they become IMBHs. Once the IMBHs reach a critical mass, they are no longer bound to stay at this region, and move inward until they inspiral into the SMBH. We aim to describe the mass distribution for the IMBHs that merge with the SMBH after such a process.

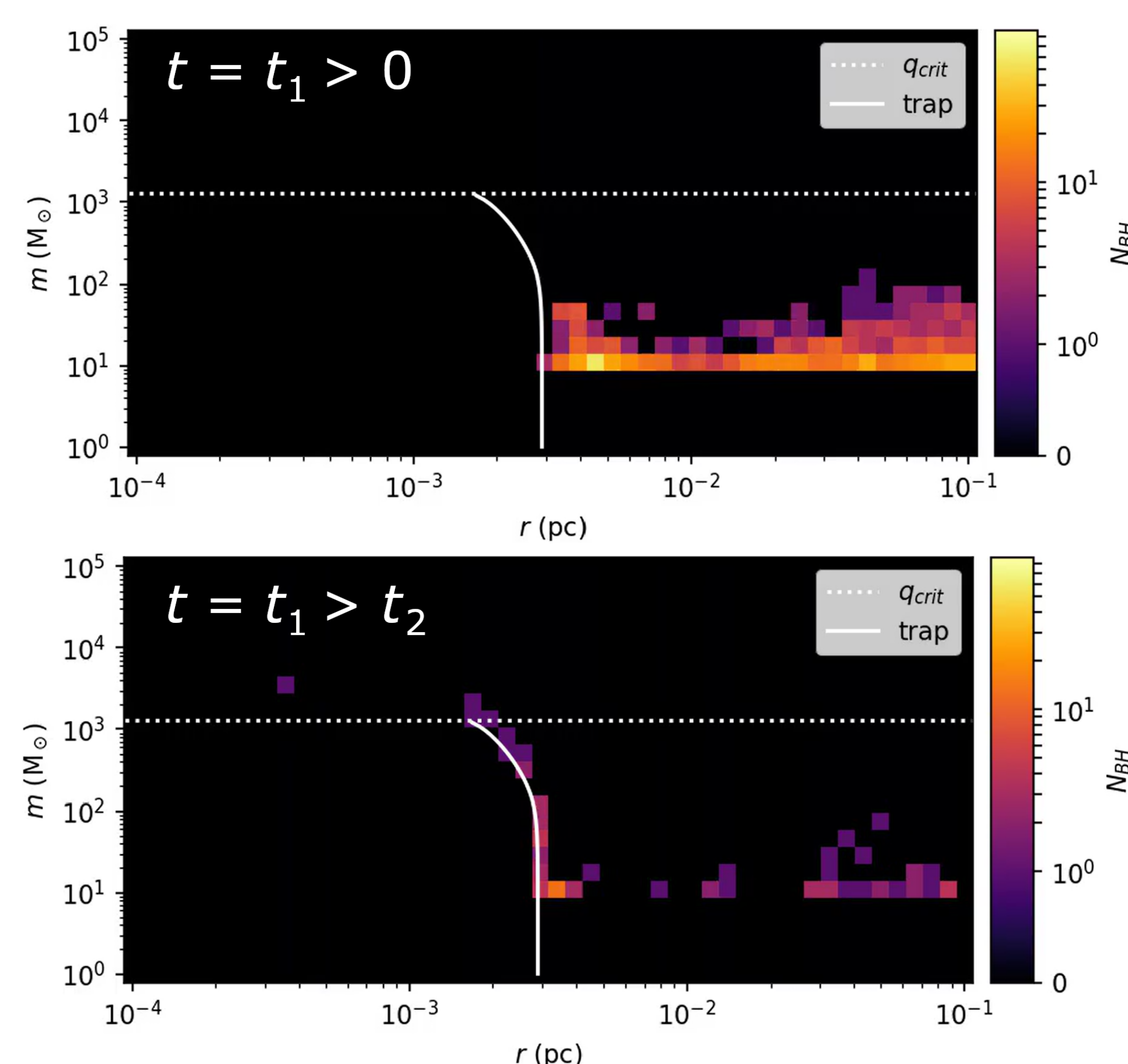


Figure 2: Two snapshots of a histogram of the simulated BHs across the m - r space (see Section 6). Solid white line: trap position for each BH mass, solved from equation (1). Dotted white line: critical mass ratio $q = m/M_{\text{SMBH}}$ where IMBHs decouple from the trap.

3. Perturbing protoplanets

Protoplanets exchange angular momentum with their PPD perturbing its surface density with respect to a PPDs without planets. Techniques to study this include [3, 4]

- fluid mechanics,
- perturbation theory, and
- expansion in Fourier modes.

There are two mechanisms for exchange of angular momentum: *Lindblad* torques relate to overdensity spirals that propagate away from the planet [3], whereas *corotation* torques relate to U-shaped streamlines ahead and behind the planet at its orbit [4, 5], as shown in Fig. 1.

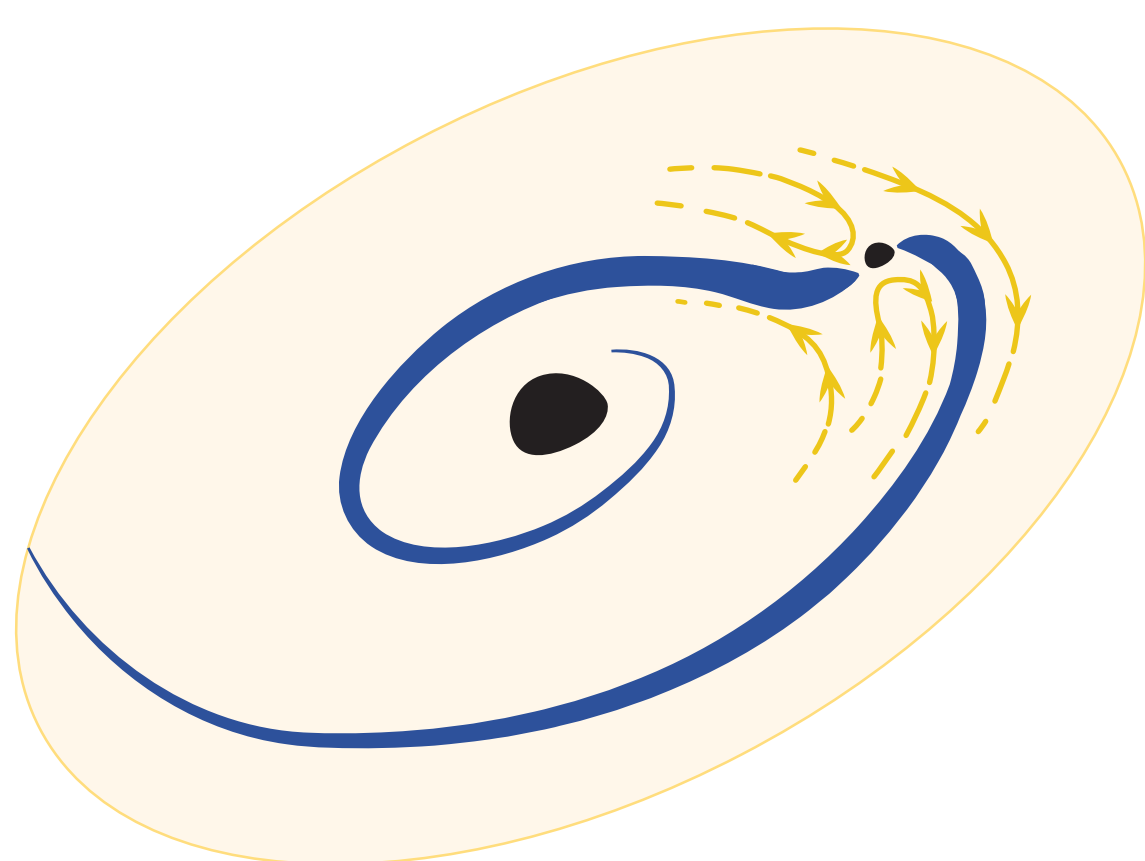


Figure 1: General picture of a PPD around a star (central black sphere) with a planet embedded in it (small black sphere). Thick blue lines: overdensity spirals. Thin yellow lines: streamlines near the orbit of the planet, the arrowheads indicate the direction of the velocity field in a rotating frame in which the planet is stationary.

4. Torque response

A planet of mass m around a star of mass M is subject to a total torque [6]

$$\Gamma_{\text{disc}} \propto \frac{\gamma_L + \gamma_c e^{-K/20}}{1 + 0.04K} \left(\frac{m}{M} \frac{H}{r} \right)^2, \quad (1)$$

where

- the Lindblad γ_L and corotation γ_c terms depend on the disc surface density $\Sigma(r)$, temperature $T(r)$, and pressure scale height $H(r)$, and
- $K = (m/M)^2 (H/r)^{-5} \alpha^{-1}$ depends on the Shakura-Sunyaev α parameter [7], characterising the viscous effects of magnetism in the disc.

5. Two contexts, same physics

We can apply this formalism to sBHs embedded in AGN discs [2]. [8] solved $\Sigma(r), T(r), H(r)$ for a typical “background” AGN disc with fiducial values $\dot{M} = c^2 L_E / 20$, $M = 10^8 M_\odot$. Key differences with a PPD include:

- [8] consider radiative support of the disc, unlike PPD models.
- AGN discs are more viscous than PPDs: $\alpha_{\text{PPD}} \sim 10^{-3}$ [9], $\alpha_{\text{AGN}} \sim 10^{-2}$ [8]. This is amplified by the exponential dependence in equation (1).
- as $r \rightarrow 0$, IMBHs lose angular momentum to GWs, $\Gamma_{\text{GW}} \propto r^{-7/2}$.

6. A tale of traps and mergers

We synthesise a population of equal-mass sBHs from a Bahcall-Wolf cusp that have been deposited on the disc [10]. We then integrate the 1-D equations of motion $\dot{r} \propto \Gamma \sqrt{r}$ during a Hubble time, considering a pair of BHs merged once they come sufficiently close to each other.

We save the masses and positions of BHs along the migration-merger process, as shown in Fig. 2. We recover migration traps and the critical mass for trap decoupling. The mass distribution of the IMBHs that reach the SMBH in less than a Hubble time is shown in Fig. 3 for different initial masses.

Further work is necessary to study the effect of non-identical initial sBH masses, as well as to sample the parameter space of critical values for the problem, such as the SMBH mass and accretion rate, and the α parameter.

References

- [1] Miralda-Escudé, & Gould. 2000, ApJ, 545, 847
- [2] Secunda, Bellovary, Mac Low, et al. 2019, ApJ, 878, 85
- [3] Goldreich, & Tremaine. 1979, ApJ, 233, 857
- [4] Paardekooper, & Papaloizou. 2008, A&A, 485, 877
- [5] Paardekooper, Baruteau, Crida, & Kley. 2010, MNRAS, 401, 1950
- [6] Kanagawa, Tanaka, & Szuszkiewicz. 2018, ApJ, 861, 140
- [7] Shakura, & Sunyaev. 1973, A&A, 24, 337
- [8] Sirko, & Goodman. 2003, MNRAS, 341, 501
- [9] Papaloizou, & Nelson. 2003, MNRAS, 339, 983
- [10] Genovozov, & Perets. 2023, MNRAS, 522, 1763